

ProfiQ

AI-Powered Employability & Career Readiness Analyzer

From Unverified Resume Claims to Observable Proof of Work — Bridging the Gap Between Student Skills and Industry Benchmarks

CORE PROJECT HIGHLIGHTS

- **Evidence-Based Skill Verification:** Cross-validates claimed technologies against live GitHub repositories, commit velocity, and code architecture rather than passive resume keywords.
- **Deterministic Readiness Scoring:** Mathematically balances skill alignment (60%) and evidence authenticity (40%) to eliminate resume inflation and gaming.
- **Actionable Career Remediation:** Diagnoses candidate role deficits and generates automated, week-by-week milestone roadmaps with curated technical resources.
- **Institutional Placement Insights:** Provides university placement cells with batch-level analytics, cohort readiness distributions, and skill gap heatmaps to run targeted campus training.

TEAM: [3Cuties]

Aman Tyagi (25BCE1710)

AI & ML Lead · Google Gemini prompt orchestration, evidence verification heuristics, schema validation & scoring engines.

Siddhant Asati (25BCE1633)

Backend & Systems Lead · Node/Express REST APIs, MongoDB data layer, GitHub telemetry extraction & service normalization.

Sahaj Kumar (25BCE1620)

Frontend & UX Lead · React 18 + Vite interactive studio, real-time readiness dials, 3D visualizers & placement dashboard.

The Employability Disconnect: Claims vs. Proof

Why traditional resumes, ATS keyword matchers, and unverified profiles fail modern engineering hiring.

01 · THE STUDENT DILEMMA

- **The Keyword Stuffing Trap:** Students list 20+ technologies on static resumes to bypass ATS filters, masking their true technical depth.
- **Fragmented Identity:** Proof of work is scattered across GitHub, LeetCode, portfolios, and courses with zero unified evaluation.
- **Reactive Rejections:** Students discover critical skill deficits only after failing technical placement rounds when it is too late to fix.

02 · THE PLACEMENT BOTTLENECK

- **Batch Review Overload:** Placement cells must evaluate 1,000+ graduating students, relying on high-level GPA rather than code readiness.
- **Zero Cohort Telemetry:** Institutions lack aggregated data to know which frameworks or technologies students are deficient in.
- **Mismatched Drives:** Students are routed to unsuitable hiring partners, resulting in poor placement conversion rates.

03 · THE RECRUITER FRICTION

- **Low Signal-to-Noise:** 70%+ of applicants with polished resumes fail basic architectural and practical coding evaluations.
- **Manual Verification Burden:** Engineering managers spend hours auditing GitHub repos, commit histories, and code quality manually.
- **Expensive Mis-Hires:** Relying on self-reported resume claims creates inflated interview funnels and high recruitment rework.

THE ROOT CAUSE: "A resume is an unverified claim. A GitHub repository is raw data."

No existing platform automatically cross-references self-reported skills against demonstrable code proof to generate an explainable, benchmarked readiness score with actionable remediation.

ProfiQ: Evidence-Based Employability Intelligence

An end-to-end AI verification engine that converts raw developer claims into verifiable career readiness.

1. INGEST & AGGREGATE

- **Resume Ingestion:** PDF parsing extracts candidate bio, education, claimed skills, and projects.
- **Developer Telemetry:** Fetches live GitHub repositories, language distributions, and commit frequency.
- **Target Role Benchmark:** Candidate selects intended role (e.g., Frontend Engineer, Full Stack, ML Engineer).

2. AI VERIFY & BENCHMARK

- **Artifact Cross-Check:** Google Gemini cross-references claimed skills against actual repo code & dependencies.
- **Confidence Scoring:** Separates verified proof from unbacked claims using deterministic heuristics.
- **Role Matrix Comparison:** Evaluates depth against standardized critical, recommended, and optional skills.

3. QUANTIFY & REMEDIATE

- **Readiness Score (0-100):** Transparent, explainable score combining skill fit (60%) and proof depth (40%).
- **Gap Identification:** Pinpoints exact missing technical dependencies required for target roles.
- **Milestone Roadmap:** Generates structured 4-week learning sprints with curated project objectives.

WHY IT IS BETTER: EVIDENCE OVER KEYWORDS

Unlike traditional ATS checkers that only perform shallow string matching, ProfiQ establishes ground truth by inspecting actual repository commits, code architecture, and package manifests to verify whether a candidate can genuinely execute.

THE INNOVATION: ACTIONABLE CAREER REMEDIATION

ProfiQ transforms evaluation from a passive 'judgment' into an active developmental roadmap. Students receive transparent explanations for every deducted point and a prescriptive milestone plan to turn red skill deficits into verified green badges.

Modular Multi-Tiered System Architecture

Engineered for high-throughput profile ingestion, robust telemetry extraction, and deterministic AI scoring.

CLIENT TIER

React 18 + Vite

- **Candidate Studio:** Interactive profile upload, role target picker, and live readiness dials.
- **Visual Telemetry:** Radar charts, verified vs. unbacked skill matrices, and 3D score dials.
- **Placement Portal:** Cohort-wide analytics, distribution heatmaps, and batch filter tables.

BACKEND API TIER

Node.js + Express

- **Ingestion Engine:** PDF parsing & multipart form handling for resumes and profile links.
- **Telemetry Service:** GitHub REST API connector extracting repo stats, languages, and commit histories.
- **Orchestrator:** Normalizes candidate profiles and manages asynchronous AI inference pipelines.

INTELLIGENCE TIER

Google Gemini + Engine

- **Skill Extraction:** Structured NER identifying explicit and implicit technical proficiencies.
- **Artifact Verifier:** Heuristics and LLM semantic cross-check matching code against claims.
- **Scoring & Roadmap:** Deterministic mathematical scoring and personalized 4-week sprint generator.

PERSISTENCE TIER

MongoDB + Schemas

- **Unified Profiles:** Stores normalized candidate claims, verified artifacts, and telemetry.
- **Role Benchmarks:** Taxonomies mapping Critical, Recommended, and Optional skills across roles.
- **Roadmaps & Reports:** Historical tracking of readiness progress over academic semesters.

KEY TECHNICAL DECISION

Modular Separation: Independent client, backend orchestrator, and AI pipelines allow replacing or upgrading LLM models without altering database or UI schemas.

HARDEST PROBLEM SOLVED

Evidence Attribution: Connecting unstructured resume claims with commit histories and repositories via semantic matching and deterministic heuristics.

VALIDATION & STABILITY

Strict JSON Schemas: Gemini structured output constraints prevent AI hallucinations, ensuring reproducible scores and robust API contracts.

Evidence-Based Mathematical Scoring Methodology

A transparent, explainable formula that rewards verified engineering competence and penalizes unbacked claims.

CORE FORMULA: Readiness Score (R) = (S × 0.60) + (E × 0.40)

Where: S = Skill Match Score (0–100) based on industry role benchmarks | E = Evidence Verification Score (0–100) based on code proof

1. SKILL MATCH SCORE (S · 60% WEIGHT)

Hierarchical Role Weights: Evaluates skills against curated industry matrices according to role criticality:

- **Critical Skills (1.0× Weight):** Foundational technologies without which the candidate cannot operate (e.g., React & JavaScript for Frontend).
- **Recommended Skills (0.7× Weight):** Modern tooling and best practices that elevate technical output (e.g., TypeScript, Tailwind, REST APIs).
- **Optional Skills (0.4× Weight):** Value-add complementary proficiencies (e.g., Docker, GraphQL, Jest).

Dynamic Breadth & Depth: Calculates coverage ratio across core domains rather than simple keyword counting.

2. EVIDENCE VERIFICATION SCORE (E · 40% WEIGHT)

Evidence Confidence Hierarchy: Each claimed skill is assigned an evidence tier backed by observable data:

- **High Confidence (0.85 – 1.0):** Public repository with verified commits, active commit velocity, and matching codebase.
- **Moderate Confidence (0.50 – 0.70):** Listed academic project, course certification, or live portfolio demo URL.
- **Low / Unbacked (0.10 – 0.30):** Resume keyword claim with zero observable codebase proof or commit history.

Penalty Mechanism: Unbacked claims dilute the overall readiness score, directly discouraging resume exaggeration.

Dual-Perspective Platform: Students & Placement Cells

Delivering immediate diagnostic value for candidate growth and powerful cohort analytics for university administrators.

CANDIDATE STUDIO: READINESS DIAL

- **Composite Readiness Score:** High-impact visual score dial reflecting current employability level (0–100).
- **Domain Radar Breakdown:** Visualizes candidate strength across Core Languages, Frameworks, Architecture, and Tooling.
- **Target Benchmark Threshold:** Clear indicators showing whether the candidate meets Junior, Mid, or Senior entry bars.

EVIDENCE MATRIX & GAP VISUALIZER

- **Verified vs. Unbacked Claims:** Displays an 80%+ evidence coverage target, clearly tagging proof sources.
- **Critical Skill Deficits:** Flags essential role requirements that are missing from candidate profiles.
- **Interactive Inspect Mode:** Allows candidates to see exactly which GitHub repos validated each skill.

PERSONALIZED MILESTONE ROADMAP

- **4-Week Sprint Schedule:** Automated sprint breakdown focusing on closing critical skill deficits step by step.
- **Curated Learning Links:** Recommends high-quality documentation, guided projects, and tutorials.
- **Proof of Work Deliverables:** Prescribes specific GitHub project milestones to generate tangible proof.

INSTITUTIONAL PLACEMENT INTELLIGENCE

- **Batch Skill Heatmaps:** Placement cells view cohort-wide trends and common technological weaknesses.
- **Pre-Placement Interventions:** Enables organizing targeted bootcamps before recruitment drives commence.
- **Verified Candidate Shortlisting:** Allows recruiters to filter students by demonstrable code proof rather than GPA alone.

Feasibility, Competitive Moat & Growth Horizon

Low operational run-rates, clear unfair advantages over legacy tools, and a defined post-hackathon roadmap.

COMPETITIVE MOAT

- **Traditional ATS:** Parses resume strings blindly; easily gamed by buzzword stuffing with zero evidence verification.
- **LinkedIn Profiles:** Relies solely on unverified self-reporting and social endorsements without code audits.
- **Generic AI Chatbots:** Provides generic advice without ingesting authentic developer artifacts or scoring mathematically.
- **The ProfiQ Advantage:** Only system combining multi-platform proof of work + industry benchmark alignment + automated remediation.

FEASIBILITY & RUN-RATE

- **Minimal API Costs:** Structured JSON prompts and token optimization keep LLM execution costs to fractions of a cent per profile.
- **Deterministic Caching:** Caches GitHub repository metadata and benchmark matrices to eliminate redundant API calls.
- **Zero Infrastructure Barrier:** Built on lightweight Node.js and MongoDB; easily deployed to serverless environments.
- **Privacy & Security:** Inspects only public code artifacts; no intrusive access to private student credentials.

FUTURE DEVELOPMENT HORIZON

- **Phase 1: Multi-Platform Ingestion:** Expand connectors to LeetCode, Codeforces, HackerRank, and GitLab for algorithmic telemetry.
- **Phase 2: University SIS Integration:** Automated batch sync with university ERPs to track student growth across semesters.
- **Phase 3: Recruiter Search Mode:** Allow tech recruiters to query candidate pools by verified skills (e.g., 'React + Docker > 85% proof').
- **Phase 4: AI Mock Interviewer:** Interactive voice agent that quizzes candidates specifically on their verified code repositories.

THANK YOU

ProfiQ — Turning Resume Claims into Verifiable Career Readiness

PROJECT VISION

- **From Claims to Proof:** Empowering engineering students to back their claims with demonstrable code.
- **Data-Driven Placement:** Giving university placement cells actionable cohort telemetry.
- **Hiring Confidence:** Providing technical recruiters with verified talent pipelines.

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- **Siddhant Asati** (25BCE1633) — Backend & Systems Lead
- **Sahaj Kumar** (25BCE1620) — Frontend & UX Lead
- **Institution:** Vellore Institute of Technology (VIT)

OPEN FOR Q&A

- **Live Demonstration:** Ready for walkthrough of the Candidate Studio & Placement Dashboard.
- **Codebase & Tests:** Modular monorepo with automated unit & end-to-end test suites.
- **Discussion:** We welcome questions on our scoring formula and verification heuristics.